

Insert Name

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1 Geometry

1.1 Geometry Primitives

```
#define EPS 1e-12
#define pi 3.1415926535897931159979634\
68544185161590576171875L
bool feq(const long double& a, const
    long double& b) {
    long double del = fabs1(a - b);
    if (del <= EPS) return true;
    return del <= EPS * max(fabs1(a),
        fabs1(b));
}
```

1.2 Point 2D

Description: Class to handle points in the plane. T can be e.g. double or long long. (Avoid int.)

```
template <class T>
int sgn(T x) {
    return (x > 0) - (x < 0);
}
template <class T>
struct Point {
    typedef Point P;
    T x, y;
    explicit Point(T x = 0, T y = 0) : x(x), y(y) {}
    bool operator<(P p) const { return tie
        (x, y) < tie(p.x, p.y); }
    bool operator==(P p) const { return
        tie(x, y) == tie(p.x, p.y); }
    P operator+(P p) const { return P(x +
        p.x, y + p.y); }
    P operator-(P p) const { return P(x -
        p.x, y - p.y); }
    P operator*(T d) const { return P(x *
        d, y * d); }
    P operator/(T d) const { return P(x /
        d, y / d); }
    T dot(P p) const { return x * p.x + y
        * p.y; }
    T cross(P p) const { return x * p.y -
        y * p.x; }
    T cross(P a, P b) const { return (a -
        *this).cross(b - *this); }
    T dist2() const { return x * x + y * y
        ; }
    double dist() const { return sqrt((
        double)dist2()); }
    // angle to x-axis in interval [-pi,
    pi]
    double angle() const { return atan2(y,
        x); }
    P unit() const { return *this / dist()
        ; } // makes dist()=1
    P perp() const { return P(-y, x); }
    // rotates +90 degrees
    P normal() const { return perp().unit
        (); }
    // returns point rotated 'a' radians
    ccw around the origin
    P rotate(double a) const {
        return P(x * cos(a) - y * sin(a), x
            * sin(a) + y * cos(a));
    }
    P project(P p) const { return *this *
        ((double)dot(p) / dist2()); }
    double scalar_project(P p) const {
        return (double)dot(p) / dist(); }
    P reflect(P p) const { return project(
        p) * 2 - p; }
    // Orientation test: >0 for left turn,
    <0 for right turn, 0 if collinear
    T side(P p1, P p2) const { return (p1
        - *this).cross(p2 - *this); }
    friend ostream& operator<<(ostream& os
        , P p) {
        return os << "(" << p.x << ", " << p.
            y << ")";
    }
```

```
}
// Same direction, length r (returns *
this if degenerate)
P truncate(double r) const { double k
    = dist(); return !sgn(k) ? *this :
    *this * (r / k); }
};
```

1.3 Line Intersection

Description:

Usage:
auto res = lineInter(s1,e1,s2,e2); if
(res.first == 1) cout << "intersection point
at " << res.second << endl;

```
template <class P>
pair<int, P> lineInter(P s1, P e1, P s2,
    P e2) {
    auto d = (e1 - s1).cross(e2 - s2);
    if (d == 0) // if parallel
        return {-(s1.cross(e1, s2) == 0), P
            (0, 0)};
    auto p = s2.cross(e1, e2), q = s2.
        cross(e2, s1);
    return {1, (s1 * p + e1 * q) / d};
}
```

1.4 Line Equation

```
array<T, 3> line_equation(const Point<T>
    &b, const Point<T>&c){
    array<T, 3>res;
    res[0] = b.y - c.y, res[1] = -(b.x - c.
        x), res[2] = b.x*c.y - b.y*c.x;
    if(res[0] < 0)res[0] = -res[0], res[1]
        = -res[1], res[2] = -res[2];
    else if(res[0] == 0 and res[1] < 0)res
        [1] = -res[1], res[2] = -res[2]; //
        for double use feq(res[0], 0)
    return res;
}
```

1.5 Segment Distance

Description:

Usage:
Point<double> a, b(2,2), p(1,1); bool
onSegment = segDist(a,b,p) < 1e-10;

```
typedef Point<double> P;
double segDist(P& s, P& e, P& p) {
    if (s == e) return (p - s).dist();
    auto d = (e - s).dist2(), t = min(d,
        max(0, (p - s).dot(e - s)));
    return ((p - s) * d - (e - s) * t).
        dist() / d;
}
```

1.6 Segment Intersection

Description:

Usage:
vector<P> inter = segInter(s1,e1,s2,e2); if
(sz(inter)==1) cout << "segments intersect at
" << inter[0] << endl;

```
template <class P>
vector<P> segInter(P a, P b, P c, P d) {
    auto oa = c.cross(d, a), ob = c.cross(
        d, b), oc = a.cross(b, c),
        od = a.cross(b, d);
    // Checks if intersection is single
    non-endpoint point.
    if (sgn(oa) * sgn(ob) < 0 && sgn(oc) *
        sgn(od) < 0)
        return {(a * ob - b * oa) / (ob - oa
            )};
    set<P> s;
    if (onSegment(c, d, a)) s.insert(a);
    if (onSegment(c, d, b)) s.insert(b);
    if (onSegment(a, b, c)) s.insert(c);
    if (onSegment(a, b, d)) s.insert(d);
    return {all(s)};
}
```

1.7 Side Of

Description:

Usage:
bool left = sideOf(p1,p2,q)==1;

```
template <class P>
int sideOf(P& s, P& e, P& p) {
    return sgn(s.cross(e, p));
}
template <class P>
int sideOf(const P& s, const P& e, const
    P& p, double eps) {
    auto a = (e - s).cross(p - s);
    double l = (e - s).dist() * eps;
    return (a > l) - (a < -l);
}
```

1.8 On Segment

Description: Returns true iff p lies on the line
segment from s to e.

Usage:
\texttt {(segDist(s,e,p)<=epsilon)} instead
when using Point<double>.

```
template <class P>
bool onSegment(P s, P e, P p) {
    return p.cross(s, e) == 0 && (s - p).
        dot(e - p) <= 0;
}
```

1.9 Linear Transformation

Description: Apply the linear transformation
(translation, rotation and scaling) which takes line
p0-p1 to line q0-q1 to point r.

```
typedef Point<double> P;
P linearTransformation(const P& p0,
    const P& p1, const P& q0, const P&
    q1,
    const P& r) {
    P dp = p1 - p0, dq = q1 - q0, num(dp.
        cross(dq), dp.dot(dq));
    return q0 + P((r - p0).cross(num), (r
        - p0).dot(num)) / dp.dist2();
}
```

1.10 Circle Line Intersection

```
template <class P>
bool circleLineIntersection(P c, double
    r, P a, P b) {
    auto d1 = (c-a).dist(), d2 = (c-b).
        dist();
    if(max(d1, d2) < r-1e-9) return false;
    ld dd;
    if ((c - a).dot(b - a) > 0 && (c - b).
        dot(a - b) > 0) {
        dd = abs((b - a).cross(c - a)) / (b
            - a).dist();
    } else {
        dd = min(d1, d2);
    }
    return dd <= r+1e-9;
}
```

1.11 Circle Intersection

```
typedef Point<double> P;
bool circleInter(P a, P b, double r1,
    double r2, pair<P, P*> out) {
    if (a == b) {
        assert(r1 != r2);
        return false;
    }
    P vec = b - a;
    double d2 = vec.dist2(), sum = r1 + r2
        , dif = r1 - r2,
        p = (d2 + r1 * r1 - r2 * r2) /
            (d2 * 2), h2 = r1 * r1 - p * p * d2
        ;
    if (sum * sum < d2 || dif * dif > d2)
        return false;
    P mid = a + vec * p, per = vec.perp()
        * sqrt(fmax(0, h2) / d2);
    *out = {mid + per, mid - per};
}
```

```
    return true;
}
```

1.12 Circle Polygon Intersection

```
typedef Point<double> P;
#define arg(p, q) atan2(p.cross(q), p.dot(q))
double circlePoly(P c, double r, vector<P> ps) {
    auto tri = [&](P p, P q) {
        auto r2 = r * r / 2;
        P d = q - p;
        auto a = d.dot(p) / d.dist2(), b = (
            p.dist2() - r * r) / d.dist2();
        auto det = a * a - b;
        if (det <= 0) return arg(p, q) * r2;
        auto s = max(0., -a - sqrt(det)), t =
            min(1., -a + sqrt(det));
        if (t < 0 || 1 <= s) return arg(p, q) * r2;
        P u = p + d * s, v = q + d * (t - 1);
        return arg(p, u) * r2 + u.cross(v) /
            2 + arg(v, q) * r2;
    };
    auto sum = 0.0;
    rep(i, 0, sz(ps)) sum += tri(ps[i] - c,
        ps[(i + 1) % sz(ps)] - c);
    return sum;
}
```

1.13 Circle Line

```
template <class P>
vector<P> circleLine(P c, double r, P a,
    P b) {
    P ab = b - a, p = a + ab * (c - a).dot(
        ab) / ab.dist2();
    double s = a.cross(b, c), h2 = r * r -
        s * s / ab.dist2();
    if (h2 < 0) return {};
    if (h2 == 0) return {p};
    P h = ab.unit() * sqrt(h2);
    return {p - h, p + h};
}
```

1.14 Circle Tangents

Description: Finds the external tangents of two circles, or internal if r2 is negated. Can return 0, 1, or 2 tangents – 0 if one circle contains the other (or overlaps it, in the internal case, or if the circles are the same); 1 if the circles are tangent to each other (in which case .first = .second and the tangent line is perpendicular to the line between the centers). .first and .second give the tangency points at circle 1 and 2 respectively. To find the tangents of a circle with a point set r2 to 0. $O(n)$

```
template <class P>
vector<pair<P, P>> tangents(P c1, double
    r1, P c2, double r2) {
    P d = c2 - c1;
    double dr = r1 - r2, d2 = d.dist2(),
        h2 = d2 - dr * dr;
    if (d2 == 0 || h2 < 0) return {};
    vector<pair<P, P>> out;
    for (double sign : {-1, 1}) {
        P v = (d * dr + d.perp() * sqrt(h2)
            * sign) / d2;
        out.push_back({c1 + v * r1, c2 + v *
            r2});
    }
    if (h2 == 0) out.pop_back();
    return out;
}
```

1.15 Minimum Enclosing Circle

Description: Computes the minimum circle that encloses a set of points. $O(n)$

```
pair<P, double> mec(vector<P> ps) {
    shuffle(all(ps), mt19937(time(0)));
    P o = ps[0];
    double r = 0, EPS = 1 + 1e-8;
    rep(i, 0, sz(ps)) if ((o - ps[i]).dist(
        ) > r * EPS) {
        o = ps[i], r = 0;
        rep(j, 0, i) if ((o - ps[j]).dist(
            ) > r * EPS) {
```

```
            o = (ps[i] + ps[j]) / 2;
            r = (o - ps[i]).dist();
            rep(k, 0, j) if ((o - ps[k]).dist(
                ) > r * EPS) {
                o = ccCenter(ps[i], ps[j], ps[k]
                );
                r = (o - ps[i]).dist();
            }
        }
    }
    return {o, r};
}
```

1.16 Inside Polygon

Description: Returns true if p lies within the polygon. If strict is true, it returns false for points on the boundary. The algorithm uses products in intermediate steps so watch out for overflow. $O(n)$

Usage:

```
vector<P> v = {P{4,4}, P{1,2}, P{2,1}}; bool
in = inPolygon(v, P{3, 3}, false);
```

```
// returns 1e9 if the point is on the
// polygon
int winding_number(const vector<P>& p,
    const P& z) { //  $O(n)$ 
    if (is_point_on_polygon(p, z)) return
        1e9;
    int n = p.size(), ans = 0;
    for (int i = 0; i < n; ++i) {
        int j = (i + 1) % n;
        bool below = p[i].y < z.y;
        if (below != (p[j].y < z.y)) {
            auto orient = z.cross(p[j], p[i]);
            if (orient == 0) return 0;
            if (below == (orient > 0)) ans +=
                below ? 1 : -1;
        }
    }
    return ans;
}
// -1 if strictly inside, 0 if on the
// polygon, 1 if strictly outside
int is_point_in_polygon(const vector<P>&
    p, const P& z) { //  $O(n)$ 
    int k = winding_number(p, z);
    return k == 1e9 ? 0 : k == 0 ? 1 : -1;
}
```

1.17 Polygon Area

Description: Returns twice the signed area of a polygon. Clockwise enumeration gives negative area. Watch out for overflow if using int as T!

```
template <class T>
T polygonArea2(vector<Point<T>>& v) {
    T a = v.back().cross(v[0]);
    rep(i, 0, sz(v) - 1) a += v[i].cross(v
        [i + 1]);
    return a;
}
```

1.18 Polygon Center

Description: Returns the center of mass for a polygon. Time: $O(n)$

```
typedef Point<double> P;
P polygonCenter(const vector<P>& v) {
    P res(0, 0);
    double A = 0;
    for (int i = 0, j = sz(v) - 1; i < sz(
        v); j = i++) {
        res = res + (v[i] + v[j]) * v[j].
            cross(v[i]);
        A += v[j].cross(v[i]);
    }
    return res / A / 3;
}
```

1.19 Polygon Cut

```
vector<P> polygonCut(const vector<P>&
    poly, P s, P e) {
    vector<P> res;
    for (int i = 0; i < (int)poly.size();
        ++i) {
        P cur = poly[i], prev = i ? poly[i -
            1] : poly.back();
```

```
        auto a = s.cross(e, cur), b = s.
            cross(e, prev);
        if ((a < 0) != (b < 0)) res.
            push_back(cur + (prev - cur) * (a /
                (a - b)));
        if (a < 0) res.push_back(cur);
    }
    return res;
}
```

1.20 Polygon Union

```
double rat(P a, P b) { return sgn(b.x) ?
    a.x / b.x : a.y / b.y; }
double polyUnion(vector<vector<P>>& poly
    ) {
    double ret = 0;
    for (int i = 0; i < (int)poly.size();
        ++i) for (int v = 0; v < (int)poly[
            i].size(); ++v) {
        P A = poly[i][v], B = poly[i][(v +
            1) % (int)poly[i].size()];
        vector<pair<double, int>> segs =
            {{0, 0}, {1, 0}};
        for (int j = 0; j < (int)poly.size()
            ; ++j) if (i != j) {
            for (int u = 0; u < (int)poly[j].
                size(); ++u) {
                P C = poly[j][u], D = poly[j][(u
                    + 1) % (int)poly[j].size()];
                int sc = sideOf(A, B, C), sd =
                    sideOf(A, B, D);
                if (sc != sd) {
                    double sa = C.cross(D, A), sb
                        = C.cross(D, B);
                    if (min(sc, sd) < 0) segs.
                        emplace_back(sa / (sa - sb), sgn(sc
                            - sd));
                } else if (!sc && !sd && j < i
                    && sgn((B - A).dot(D - C)) > 0) {
                    segs.emplace_back(rat(C - A, B
                        - A), 1);
                    segs.emplace_back(rat(D - A, B
                        - A), -1);
                }
            }
        }
        sort(segs.begin(), segs.end());
        for (auto& s : segs) s.first = min(
            max(s.first, 0.0), 1.0);
        double sum = 0;
        int cnt = segs[0].second;
        for (int j = 1; j < (int)segs.size()
            ; ++j) {
            if (!cnt) sum += segs[j].first -
                segs[j - 1].first;
            cnt += segs[j].second;
        }
        ret += A.cross(B) * sum;
    }
    return ret / 2;
}
```

1.21 Extreme Vertex

```
// id of the vertex having maximum dot
// product with z
// polygon must be convex
// top - upper right vertex
// for minimum dot product negate z and
// return -p[id].dot(z)
int extreme_vertex(const vector<P>& p,
    const P& z, const int top) { //  $O(
        \log n)$ 
    int n = p.size();
    if (n == 1) return 0;
    auto ans = p[0].dot(z);
    int id = 0;
    if (p[top].dot(z) > ans) ans = p[top].
        dot(z), id = top;
    int l = 1, r = top - 1;
    while (l < r) {
        int mid = (l + r) / 2;
        if (p[mid + 1].dot(z) >= p[mid].dot(
            z)) l = mid + 1;
        else r = mid;
    }
    if (p[l].dot(z) > ans) ans = p[l].dot(
        z), id = l;
```

```

l = top + 1, r = n - 1;
while (l < r) {
    int mid = (l + r) / 2;
    if (p[(mid + 1) % n].dot(z) >= p[mid].dot(z)) l = mid + 1;
    else r = mid;
}
l %= n;
if (p[l].dot(z) > ans) ans = p[l].dot(z), id = l;
return id;
}

```

1.22 Manhattan MST

```

vector<array<int, 3>> manhattanMST(
    vector<P> ps) {
    vector<int> id(ps.size());
    iota(id.begin(), id.end(), 0);
    vector<array<int, 3>> edges;
    for (int k = 0; k < 4; ++k) {
        sort(id.begin(), id.end(),
            [&](int i, int j) { return (ps[i] - ps[j]).x < (ps[j] - ps[i]).y; });
        map<int, int> sweep;
        for (int i : id) {
            for (auto it = sweep.lower_bound(-ps[i].y); it != sweep.end(); sweep.erase(it++)) {
                int j = it->second;
                P d = ps[i] - ps[j];
                if (d.y > d.x) break;
                edges.push_back({d.y + d.x, i, j});
            }
            sweep[-ps[i].y] = i;
        }
        for (P& p : ps)
            if (k & 1)
                p.x = -p.x;
            else
                swap(p.x, p.y);
    }
    return edges;
}

```

1.23 Convex Hull

Description: Returns a vector of the points of the convex hull in counter-clockwise order. Points on the edge of the hull between two other points are not considered part of the hull. $O(n \log n)$

```

typedef Point<ll> P;
vector<P> convexHull(vector<P> pts) {
    int n = pts.size();
    if (n <= 1) return pts;
    sort(all(pts));
    vector<P> h(n + 1);
    int s = 0, t = 0;
    for (int it = 2; it--; s = --t, reverse(all(pts)))
        for (P p : pts) {
            while (t >= s + 2 && h[t - 2].cross(h[t - 1], p) <= 0) t--;
            h[t++] = p;
        }
    return {h.begin(), h.begin() + t - (t == 2 && h[0] == h[1])};
}

```

1.24 Hull Diameter

Description: Returns the two points with max distance on a convex hull (ccw, no duplicate/collinear points). $O(n)$

```

typedef Point<ll> P;
array<P, 2> hullDiameter(vector<P> S) {
    int n = sz(S), j = n < 2 ? 0 : 1;
    pair<ll, array<P, 2>> res({0, {S[0], S[0]}});
    rep(i, 0, j) for (; j = (j + 1) % n) {
        res = max(res, {(S[i] - S[j]).dist2(), {S[i], S[j]}});
        if ((S[(j + 1) % n] - S[j]).cross(S[i + 1] - S[i]) >= 0) break;
    }
    return res.second;
}

```

```

}

```

1.25 Point Inside Hull

Description: Determine whether a point t lies inside a convex hull (CCW order, with no collinear points). Returns true if point lies within the hull. If strict is true, points on the boundary aren't included. $O(\log N)$

```

typedef Point<ll> P;
bool inHull(vector<P>& l, P p, bool strict = true) {
    int n = l.size(), a = 1, b = n - 1, r = !strict;
    if (n < 3) return r && onSegment(l[0], l.back(), p);
    if (sideOf(l[0], l[a], l[b]) > 0) swap(a, b);
    if (sideOf(l[0], l[a], p) >= r || sideOf(l[0], l[b], p) <= -r) return false;
    while (abs(a - b) > 1) {
        int c = (a + b) / 2;
        if (sideOf(l[0], l[c], p) > 0 ? b : a) = c;
    }
    return sgn(l[a].cross(l[b], p)) < r;
}

```

1.26 Line Hull Intersection

Description: Line-convex polygon intersection. The polygon must be ccw and have no collinear points. `lineHull(line, poly)` returns a pair describing the intersection of a line with the polygon:

- $(-1, -1)$ if no collision,
- $(i, -1)$ if touching the corner i ,
- (i, i) if along side $(i, i + 1)$,
- (i, j) if crossing sides $(i, i + 1)$ and $(j, j + 1)$.

In the last case, if a corner i is crossed, this is treated as happening on side $(i, i + 1)$. The points are returned in the same order as the line hits the polygon. `extrVertex` returns the point of a hull with the max projection onto a line. $O(\log N)$

```

#define cmp(i, j) sgn(dir.perp().cross(poly[(i) % n] - poly[(j) % n]))
#define extr(i) cmp(i + 1, i) >= 0 && cmp(i, i - 1 + n) < 0
template <class P>
int extrVertex(vector<P>& poly, P dir) {
    int n = sz(poly), lo = 0, hi = n;
    if (extr(0)) return 0;
    while (lo + 1 < hi) {
        int m = (lo + hi) / 2;
        if (extr(m)) return m;
        int ls = cmp(lo + 1, lo), ms = cmp(m + 1, m);
        (ls < ms || (ls == ms && ls == cmp(lo, m)) ? hi : lo) = m;
    }
    return lo;
}
#define cmpL(i) sgn(a.cross(poly[i], b))
template <class P>
array<int, 2> lineHull(P a, P b, vector<P>& poly) {
    int endA = extrVertex(poly, (a - b).perp());
    int endB = extrVertex(poly, (b - a).perp());
    if (cmpL(endA) < 0 || cmpL(endB) > 0) return {-1, -1};
    array<int, 2> res;
    rep(i, 0, 2) {
        int lo = endB, hi = endA, n = sz(poly);
        while ((lo + 1) % n != hi) {
            int m = ((lo + hi + (lo < hi ? 0 : n)) / 2) % n;
            (cmpL(m) == cmpL(endB) ? lo : hi) = m;
        }
        res[i] = (lo + !cmpL(hi)) % n;
        swap(endA, endB);
    }
    if (res[0] == res[1]) return {res[0], -1};
    if (!cmpL(res[0]) && !cmpL(res[1]))

```

```

switch ((res[0] - res[1] + sz(poly) + 1) % sz(poly)) {
    case 0:
        return {res[0], res[0]};
    case 2:
        return {res[1], res[1]};
}
return res;
}

```

1.27 Half Plane Intersection

```

typedef Point<double> P;
using Line = pair<P, P>;
bool bad(Line a, Line b, Line c) {
    auto res = lineInter(a.first, a.second, b.first, b.second);
    return res.first > 0 && sideOf(c.first, c.second, res.second, EPS) < 0;
}
vector<P> halfPlaneIntersection(vector<Line>& L) {
    auto ang = [](Line l) { return (l.second - l.first).angle(); };
    sort(all(L), [&](Line a, Line b) { return ang(a) < ang(b); });
    vector<Line> lines;
    for (int i = 0; i < L.size(); i++) {
        if (i && abs(ang(L[i]) - ang(L[i - 1])) < EPS) {
            if (sideOf(L[i].first, L[i].second, lines.back().first, EPS) < 0) {
                lines.pop_back(); lines.push_back(L[i]);
            }
            else lines.push_back(L[i]);
        }
        int n = lines.size(), qf = 0, qb = 0;
        vector<Line> q(n);
        for (int i = 0; i < n; i++) {
            while (qb - qf >= 2 && bad(q[qb - 1], q[qb - 2], lines[i])) qb--;
            while (qb - qf >= 2 && bad(q[qf], q[qf + 1], lines[i])) qf++;
            q[qb++] = lines[i];
        }
        while (qb - qf >= 3 && bad(q[qb - 1], q[qb - 2], q[qf])) qb--;
        while (qb - qf >= 3 && bad(q[qf], q[qf + 1], q[qb - 1])) qf++;
        if (qb - qf < 3) return {};
        vector<P> poly;
        for (int i = qf; i < qb; i++) {
            int nxt = (i + 1 == qb) ? qf : i + 1;
            poly.push_back(lineInter(q[i].first, q[i].second, q[nxt].first, q[nxt].second).second);
        }
        return poly;
    }
}

```

1.28 Minkowski Sum Dif

```

using P = Point<ld>;
void reorder(vector<P>& poly) {
    int pos = 0;
    for (int i = 1; i < (int)poly.size(); ++i)
        if (poly[i] < poly[pos]) pos = i;
    rotate(poly.begin(), poly.begin() + pos, poly.end());
}
vector<P> minkowski(vector<P> a, vector<P> b) {
    reorder(a), reorder(b);
    int n = a.size(), m = b.size(), ai = 0, bi = 0;
    a.push_back(a[0]), a.push_back(a[1]);
    b.push_back(b[0]), b.push_back(b[1]);
    vector<P> c;
    while (ai < n || bi < m) {
        c.push_back(a[ai] + b[bi]);
        if (ai < n && bi < m) {
            auto ori = (a[ai + 1] - a[ai]).cross(b[bi + 1] - b[bi]);
            if (ori > 1e-9) ai++;
            else if (ori < -1e-9) bi++;
            else ai++, bi++;
        }
        else if (ai < n) ai++;
    }
}

```

```

    else bi++;
}
return c;
}
vector<P> minkowski_dif(vector<P> a,
    vector<P> b) {
    for (auto& p : b) p.x = -p.x, p.y = -p.y;
    return minkowski(a, b);
}

```

1.29 Maximum Inscribed Circle

```

ld maxInscribedCircle(vector<P>& poly) {
    ld lo = 0, hi = 1e6, ans = 0, L = 1e9;
    int n = poly.size();
    while (hi - lo >= 1e-12) {
        ld mid = lo + (hi - lo) / 2.0;
        vector<Line> l = {{-L, -L}, {L, -L}}, {{L, -L}, {L, L}}, {{L, L}, {-L, L}}, {{-L, L}, {-L, -L}};
        for (int i = 0; i < n; ++i) {
            auto a = poly[i], b = poly[(i + 1) % n];
            auto c = (b - a).unit().perp();
            l.push_back({a + c * mid, b + c * mid});
        }
        auto cur = halfPlaneIntersection(l);
        if (!cur.empty()) ans = lo = mid;
        else hi = mid;
    }
    return ans;
}

```

1.30 Rotating Calipers

```

// Outer loop advances edge (i -> i+1),
// inner loop advances antipodal pointer j
for (int i = 0, j = 1; i < n; ++i) {
    while ((p[(i + 1) % n] - p[i]).cross(p[(j + 1) % n] - p[j]) > 0) {
        j = (j + 1) % n;
    }
    // Process antipodal pair: p[i] (or edge i -> i+1) and p[j]
}

```

1.31 Closest Pair

Description: Finds the closest pair of points. $O(n \log n)$

```

typedef Point<ll> P;
pair<P, P> closest(vector<P> v) {
    assert(sz(v) > 1);
    set<P> S;
    sort(all(v), [](P a, P b) { return a.y < b.y; });
    pair<ll, pair<P, P>> ret{LLONG_MAX, {P(), P()}};
    int j = 0;
    for (P p : v) {
        P d{1 + (ll)sqrt(ret.first), 0};
        while (v[j].y <= p.y - d.x) S.erase(v[j++]);
        auto lo = S.lower_bound(p - d), hi = S.upper_bound(p + d);
        for (; lo != hi; ++lo) ret = min(ret, {(ll)*lo - p).dist2(), {(ll)*lo, p}});
        S.insert(p);
    }
    return ret.second;
}

```

2 Geometry 3D

2.1 Point 3D

```

using P2 = Point<double>;
struct P3 {
    double x, y, z;
    explicit P3(double x = 0, double y = 0, double z = 0) : x(x), y(y), z(z) {}
    bool operator<(P3 p) const { return tie(x, y, z) < tie(p.x, p.y, p.z); }
}

```

```

bool operator==(P3 p) const { return tie(x, y, z) == tie(p.x, p.y, p.z); }
P3 operator+(P3 p) const { return P3(x + p.x, y + p.y, z + p.z); }
P3 operator-(P3 p) const { return P3(x - p.x, y - p.y, z - p.z); }
P3 operator*(double d) const { return P3(x * d, y * d, z * d); }
P3 operator/(double d) const { return P3(x / d, y / d, z / d); }
double dot(P3 p) const { return x * p.x + y * p.y + z * p.z; }
P3 cross(P3 p) const { return P3(y * p.z - z * p.y, z * p.x - x * p.z, x * p.y - y * p.x); }
double dist2() const { return x * x + y * y + z * z; }
double dist2(P3 p) const { return (*this - p).dist2(); }
double dist() const { return sqrt(dist2()); }
double dist(P3 p) const { return (*this - p).dist(); }
P3 unit() const { return *this / dist(); }
double angle(P3 p) const { return atan2(cross(p).dist(), dot(p)); }
// (this*a).b
double triple(P3 a, P3 b) const { return cross(a).dot(b); }
// Smallest angle between this and p, ignoring direction, in [0, PI/2]
double small_angle(P3 p) const { return acos(min(fabs(dot(p)) / dist(), 1.0)); }
// Same direction, length r
P3 truncate(double r) const { double k = dist(); return !sgn(k) ? *this : *this * (r / k); }
// Rotate vector around unit axis 'u' by angle 'a' (Rodrigues' rotation formula)
P3 rotate(P3 u, double a) const { u = u.unit(); return *this * cos(a) + u.cross(*this) * sin(a) + u * u.dot(*this) * (1 - cos(a)); }
// Returns an arbitrary orthogonal unit vector to this
P3 perp() const { return fabs(x) > EPS || fabs(y) > EPS ? P3(-y, x, 0).unit() : P3(0, -z, y).unit(); }
};

```

```

// orient' is >0 if s lies on the same side of plane pqr as (q-p) x (r-p), <0 opposite, 0 on it.
// orient_by_normal: 2D-like orientation of p->q->r on the plane with normal n.
double orient(P3 p, P3 q, P3 r, P3 s) { return (q - p).cross(r - p).dot(s - p); }
double orient_by_normal(P3 p, P3 q, P3 r, P3 n) { return (q - p).cross(r - p).dot(n); }

```

2.2 Plane

```

// 3D Plane: normal n, offset d (n.dot(p) = d)
struct Plane {
    P3 n; double d;
    Plane(P3 n, double d) : n(n), d(d) {}
    Plane(P3 n, P3 p) : n(n), d(n.dot(p)) {}
    Plane(P3 p, P3 q, P3 r) : Plane((q - p).cross(r - p), p) {}
    double side(P3 p) const { return n.dot(p) - d; }
    double dist(P3 p) const { return fabs(side(p)) / n.dist(); }
    P3 proj(P3 p) const { return p - n * (side(p) / n.dist2()); }
    P3 reflect(P3 p) const { return p - n * (2 * side(p) / n.dist2()); }
    // [-PI, PI]

```

```

double angle(Plane p) const { return n.angle(p.n); }
// [0, PI/2]
double small_angle(Plane p) const { return n.small_angle(p.n); }
bool is_para(Plane p) const { return n.cross(p.n).dist() < EPS; }
bool is_perp(Plane p) const { return sgn(n.dot(p.n)) == 0; }
// Translate the plane by vector t
Plane translate(P3 t) const { return {n, d + n.dot(t)}; }
// Shift the plane along n by distance dis
Plane shift_up(double dis) const { return {n, d + dis * n.dist()}; }
// Two distinct points lying on the plane (n must be non-zero)
pair<P3, P3> two_pts() const { int sx = sgn(n.x), sy = sgn(n.y), sz = sgn(n.z); assert(sx || sy || sz); if (!sx && !sy) return {P3(1, 0, d / n.z), P3(0, 1, d / n.z)}; if (!sy && !sz) return {P3(d / n.x, 1, 0), P3(d / n.x, 0, 1)}; if (!sz && !sx) return {P3(1, d / n.y, 0), P3(0, 1, d / n.y, 1)}; if (!sx) return {P3(1, d / n.y, 0), P3(0, 0, d / n.z)}; if (!sy) return {P3(0, 1, d / n.z), P3(d / n.x, 0, 0)}; if (!sz) return {P3(d / n.x, 0, 1), P3(0, d / n.y, 0)}; if (sgn(d)) return {P3(d / n.x, 0, 0), P3(0, d / n.y, 0)}; return {P3(n.y, -n.x, 0), P3(-n.y, n.x, 0)}; }
};
// Intersection line of two planes 3-point pairs.
pair<P3, P3> plane_inter(P3 a1, P3 b1, P3 c1, P3 a2, P3 b2, P3 c2) { auto n1 = (b1 - a1).cross(c1 - a1), n2 = (b2 - a2).cross(c2 - a2); auto d = n1.cross(n2); if (d.dist2() == 0) return {P3(), P3()}; auto o = (n2 * n1.dot(a1) - n1 * n2.dot(a2)).cross(d) / d.dist2(); return {o, o + d}; }

```

2.3 Line 3D

```

// 3d line parametric form: 'o + k * d'.
struct Line3D {
    P3 d, o;
    Line3D(P3 p, P3 q) : d(q - p), o(p) {}
    Line3D(Plane p1, Plane p2) { d = p1.n.cross(p2.n); o = (p2.n * p1.d - p1.n * p2.d).cross(d) / d.dist2(); }
    double dist2(P3 p) const { return d.cross(p - o).dist2() / d.dist2(); }
    double dist(P3 p) const { return sqrt(dist2(p)); }
    P3 proj(P3 p) const { return o + d * (d.dot(p - o) / d.dist2()); }
    P3 inter(Plane p) const { return o - d * (p.side(o) / d.dot(p.n)); }
    // Orthogonal reflection of point p across the line
    P3 reflect(P3 p) const { return proj(p) * 2 - p; }
    // Compare points by their projection on the line (use as a sort comparator)
    bool cmp_proj(P3 p, P3 q) const { return d.dot(p) < d.dot(q); }
    // Small angle between the direction vectors of two lines
    double angle(Line3D l) const { return d.small_angle(l.d); }
    bool is_para(Line3D l) const { return

```

```

    d.cross(l.d).dist() < EPS; }
bool is_perp(Line3D l) const { return
    sgn(d.dot(l.d)) == 0; }
// Distance from 3D segment [p, q] to
// point x
static double seg_dist(P3 p, P3 q, P3
    x) {
    if ((q - p).dot(x - p) < 0) return x
        .dist(p);
    if ((p - q).dot(x - q) < 0) return x
        .dist(q);
    return (q - p).unit().cross(x - p).
        dist();
}
// Check if point p lies on segment
// defined by [o, o + d]
bool on_segment(P3 p) const {
    return dist(p) < EPS && (o - p).dot(
        o + d - p) <= EPS;
}
};
// Line-line relations: min dist,
// closest point
double line_dist(Line3D l1, Line3D l2) {
    auto n = l1.d.cross(l2.d);
    if (n.dist2() == 0) return l1.dist(l2.
        o);
    return fabs((l2.o - l1.o).dot(n)) / n.
        dist();
}
P3 line_closest(Line3D l1, Line3D l2) {
    auto n2 = l2.d.cross(l1.d.cross(l2.d))
        ;
    return l1.o + l1.d * ((l2.o - l1.o).
        dot(n2) / l1.d.dot(n2));
}
}

```

2.4 Plane Line

```

// Plane-line angles, relations, and
// perpendiculars through a point.
double angle(Plane p, Line3D l) { return
    PI / 2 - p.n.small_angle(l.d); }
bool is_para(Plane p, Line3D l) { return
    sgn(p.n.dot(l.d)) == 0; }
bool is_perp(Plane p, Line3D l) { return
    p.n.cross(l.d).dist() < EPS; }
Line3D perp_through(Plane p, P3 o) {
    return Line3D(o, o + p.n); }
Plane perp_through(Line3D l, P3 o) {
    return Plane(l.d, o); }

```

2.5 Convex Hull 3d

```

// O(N^2) Incremental 3D Convex Hull
// builder (outputs flattened triangle
// vector)
struct CH3D {
    struct face { int a, b, c; bool ok; };
    int n, num;
    vector<P3> P;
    vector<face> F;
    vector<vector<int>> g;
    void init(vector<P3> p) {
        P = p; n = p.size(); num = 0;
        F.resize(8 * n + 1); g.assign(n + 1,
            vector<int>(n + 1, 0));
    }
    double volume(P3 a, P3 b, P3 c, P3 d)
        { return ((b - a).cross(c - a)).dot
            (d - a); }
    double dblcmp(P3 &p, face &f) { return
        ((P[f.b] - P[f.a]).cross(P[f.c] -
            P[f.a])).dot(p - P[f.a]); }
    void deal(int p, int a, int b) {
        int f = g[a][b]; face add;
        if (F[f].ok) {
            if (dblcmp(P[p], F[f]) > EPS) dfs(
                p, f);
            else {
                add = {b, a, p, true};
                g[p][b] = g[a][p] = g[b][a] =
                    num;
                F[num++] = add;
            }
        }
    }
    void dfs(int p, int now) {
        F[now].ok = false;

```

```

        deal(p, F[now].b, F[now].a);
        deal(p, F[now].c, F[now].b);
        deal(p, F[now].a, F[now].c);
    }
    void build() {
        if (n < 4) return;
        bool ok = false;
        for (int i = 1; i < n; i++) if (P
            [0].dist(P[i]) > EPS) { swap(P[1],
                P[i]); ok = true; break; }
        if (!ok) return; ok = false;
        for (int i = 2; i < n; i++) if ((P
            [0] - P[i]).cross(P[1] - P[i]).dist
                () > EPS) { swap(P[2], P[i]); ok =
                    true; break; }
        if (!ok) return; ok = false;
        for (int i = 3; i < n; i++) if (fabs
            (((P[0] - P[i]).cross(P[1] - P[2]))
                .dot(P[0] - P[i])) > EPS) { swap(P
                [3], P[i]); ok = true; break; }
        if (!ok) return;
        for (int i = 0; i < 4; i++) {
            face add = {(i + 1) % 4, (i + 2) %
                4, (i + 3) % 4, true};
            if (dblcmp(P[i], add) > 0) swap(
                add.b, add.c);
            g[add.a][add.b] = g[add.b][add.c]
                = g[add.c][add.a] = num;
            F[num++] = add;
        }
        for (int i = 4; i < n; i++) {
            for (int j = 0; j < num; j++) if (
                F[j].ok && dblcmp(P[i], F[j]) > EPS
                ) { dfs(i, j); break; }
        }
        int tmp = num; num = 0;
        for (int i = 0; i < tmp; i++) if (F[
            i].ok) F[num++] = F[i];
    }
    // center of mass
    P3 centroid() {
        P3 ans(0, 0, 0), o(0, 0, 0); double
            all = 0;
        for (int i = 0; i < num; i++) {
            double vol = volume(o, P[F[i].a],
                P[F[i].b], P[F[i].c]);
            ans = ans + (o + P[F[i].a] + P[F[i
                ].b] + P[F[i].c]) / 4.0 * vol;
            all += vol;
        }
        return ans / all;
    }
    // Total surface area of the 3D Convex
    // Hull
    double area() {
        if (n == 3) return (P[1] - P[0]).
            cross(P[2] - P[0]).dist() * 0.5;
        double res = 0;
        for (int i = 0; i < num; i++) {
            res += (P[F[i].b] - P[F[i].a]).
                cross(P[F[i].c] - P[F[i].a]).dist()
                ;
        }
        return res * 0.5;
    }
    // Total volume enclosed by the 3D
    // Convex Hull
    double volume() {
        double res = 0; P3 o(0, 0, 0);
        for (int i = 0; i < num; i++) res +=
            volume(o, P[F[i].a], P[F[i].b], P[
                F[i].c]);
        return fabs(res / 6.0);
    }
    // Are the triangular faces s and t
    // coplanar?
    bool same(int s, int t) {
        P3 a = P[F[s].a], b = P[F[s].b], c =
            P[F[s].c];
        return fabs(volume(a, b, c, P[F[t].a
            ]) < EPS && fabs(volume(a, b, c, P
                [F[t].b])) < EPS &&
            fabs(volume(a, b, c, P[F[t].c
            ]) < EPS;
    }
    int num_tri() { return num; } //
    // number of surface triangles
    int num_poly() { //

```

```

        // number of surface polygons, 0(num^2)
        int res = 0;
        for (int i = 0; i < num; i++) {
            bool ok = true;
            for (int j = 0; j < i; j++) if (
                same(i, j)) { ok = false; break; }
            res += ok;
        }
        return res;
    }
    // Distance from point p to the plane
    // of face i
    double pt_face_dist(P3 p, int i) {
        return fabs(volume(P[F[i].a], P[F[i
            ].b], P[F[i].c], p)) /
            ((P[F[i].b] - P[F[i].a]).cross
                (P[F[i].c] - P[F[i].a]).dist());
    }
    // Check if point p is inside or on
    // the 3D Convex Hull
    bool contains(P3 p) {
        for (int i = 0; i < num; i++) {
            if (dblcmp(p, F[i]) > EPS) return
                false;
        }
        return true;
    }
};

```

2.6 Sphere

```

// Spherical coordinates, distances,
// and sphere-sphere intersections.
P3 sph_pt(double r, double lat, double
    lon) {
    lat *= PI / 180; lon *= PI / 180;
    return P3(r * cos(lat) * cos(lon), r *
        cos(lat) * sin(lon), r * sin(lat))
        ;
}
double great_circle_dist(P3 o, double r,
    P3 a, P3 b) { return r * (a - o).
    angle(b - o); }
struct Sphere {
    P3 c; double r;
    double volume() const { return 4.0 /
        3.0 * PI * r * r * r; }
    double surface_area() const { return 4
        * PI * r * r; }
    // Line-Sphere Intersection Points
    vector<P3> inter(Line3D l) const {
        P3 proj = l.proj(c);
        double d2 = c.dist2(proj);
        if (d2 > r * r + EPS) return {};
        double h = sqrt(max(0.0, r * r - d2)
            );
        P3 dir = l.d.unit();
        if (h < EPS) return {proj};
        return {proj - dir * h, proj + dir *
            h};
    }
    // if this point is inside the sphere
    bool contains(P3 p) const { return sgn
        (p.dist(c) - r) <= 0; }
    // Length of the tangent segment from
    // an external point p (-1 if p is
    // inside)
    double tangent_len(P3 p) const {
        double d = p.dist(c);
        return sgn(d - r) < 0 ? -1 : sqrt(
            max(0.0, d * d - r * r));
    }
    // Polar angle of the cap of tangency
    // points from p (-1 if inside, -2 if
    // on the surface)
    double tangent_angle(P3 p) const {
        double d = p.dist(c);
        if (sgn(d - r) < 0) return -1;
        if (sgn(d - r) == 0) return -2;
        return acos(r / d);
    }
};
// sphere-sphere intersection
pair<double, double> sphere_inter(Sphere
    s1, Sphere s2) {
    double d = s1.c.dist(s2.c);
    if (d >= s1.r + s2.r) return {0, 0};
    double R = max(s1.r, s2.r), r = min(s1

```

```

    .r, s2.r);
double x = (R * R - r * r + d * d) /
(2 * d);
double h1 = R - x, h2 = (R + r - d) -
h1;
double vol = PI * h1 * h1 * (3 * R -
h1) / 3.0 + PI * h2 * h2 * (3 * r -
h2) / 3.0;
double area = 2 * PI * R * h1 + 2 * PI
* r * h2;
return {area, vol};
}
// Radius of the circle in which two
// intersecting spheres meet
double sphere_inter_radius(Sphere s1,
Sphere s2) {
double d = s1.c.dist(s2.c), R = max(s1
.r, s2.r), r = min(s1.r, s2.r);
if (sgn(d - R - r) >= 0) return 0;
double w = d * d - r * r + R * R;
return sqrt(max(0.0, 4 * d * d * R * R
- w * w)) / (2 * d);
}
// Sphere through four non-coplanar
// points
Sphere circum_sphere(P3 a, P3 b, P3 c,
P3 d) {
assert(sgn(Plane(a, b, c).side(d)) !=
0);
Plane u(a - b, (a + b) / 2), v(b - c,
(b + c) / 2), w(c - d, (c + d) / 2)
;
Line3D l1(u, v), l2(v, w);
P3 o = line_closest(l1, l2);
return {o, o.dist(a)};
}
// Center of the circle through three
// non-collinear coplanar points.
P3 circle_center(P3 a, P3 b, P3 c) {
P3 v1 = b - a, v2 = c - a;
double v11 = v1.dot(v1), v22 = v2.dot(
v2), v12 = v1.dot(v2);
double base = 0.5 / (v11 * v22 - v12 *
v12);
return a + v1 * (base * v22 * (v11 -
v12)) + v2 * (base * v11 * (v22 -
v12));
}
// approximate min enclosing sphere
Sphere min_sphere(vector<P3> p) {
P3 c(0, 0, 0); int n = p.size(), pos =
0;
for (auto &i : p) c = c + i;
c = c / n;
double ratio = 0.1;
for (int it = 0; it < 100000; it++) {
pos = 0;
for (int i = 1; i < n; i++) if (c.
dist2(p[i]) > c.dist2(p[pos])) pos
= i;
c = c + (p[pos] - c) * ratio;
ratio *= 0.998;
}
return {c, c.dist(p[pos])};
}

```